Two forbidden patterns bound the boundary intersections of two simple heptagons

A Lean-verified proof that $|\partial P \cap \partial Q| \leq 38$ when no vertex lies on the other boundary — the case $f(7,7) = 38$ assuming a general-position reduction we do not prove

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Abstract

Let $f(m,n)$ be the maximum number of points at which the boundaries of two simple polygons with m and n sides can meet. Dillencourt, Mount and Saalfeld posed the problem in 1993 and settled the cases where m or n is even; the odd–odd cases have resisted, and $m = n = 7$ is the smallest open case. We prove that two simple heptagons, neither having a vertex on the other's boundary, meet in at most 38 boundary points, and that 38 is attained. The hypothesis on vertices is a published reduction we import rather than prove, so this is the case $f(7,7) = 38$ *conditional* on that import, not a resolution of it.

The mathematical content is a reduction. Rather than enumerate the 13,003,620 candidate crossing patterns with 40 intersections — the number of 9-subsets of the 7×7 grid meeting every row and column, or about 33,000 after quotienting by symmetry — we exhibit *two* local patterns, prescribing 8 and 14 side-pair incidences respectively (the second together with seven self-non-crossing conditions), and show that every candidate pattern with 40 intersections and every one with 42 contains one of them. Each of the two is non-realizable, and a degree argument over 28 cases, itself discharged by machine, reduces the whole exclusion to those two facts. That discharge is not hand-checkable (Remark 4.2).

Every mathematical assertion in §§2–7 is verified by the Lean 4 kernel from primitive definitions, with no published result about $f(m,n)$ appealed to. Two negative observations in §8.1 are analytic and are *not* machine-checked. The statement we do not prove is the reduction to general position, for which we import Ackerman–Keszegh–Rote §9 and carry it as an explicit hypothesis. We are explicit throughout about which assertions are machine-checked, which are imported, and which — notably the faithfulness of our definitions — remain matters for a human reader.

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1 Introduction

1.1 The problem

Two simple polygons in the plane, with m and n sides, are in *general position* if no vertex of either lies on the boundary of the other and no three of the combined $m + n$ vertices are collinear. Write $f(m, n)$ for the maximum, over such pairs, of $|\partial P \cap \partial Q|$.

Problem (Dillencourt–Mount–Saalfeld [6]). Determine $f(m, n)$.

They determined $f(m, n)$ when at least one of m, n is even. The odd–odd cases are open. A 2012 preprint of Günther [7] claiming $f(n, n)$ for odd n was withdrawn by its author, the stated reason being "an unrecoverable error in the proof of Lemma 2", with a counterexample found to a weaker form of that lemma. The smallest open case is $m = n = 7$.

Remark 1.1 (A caution for anyone searching the literature). Several web search engines, asked about this problem, return summaries asserting that the odd–odd conjecture *is* proved. Those summaries paraphrase the *withdrawn* preprint. We mention it because we were shown such a summary during preparation.

Two elementary facts frame it. Each side of P meets each side of Q at most once in general position, so $I \leq 49$; and since ∂P and ∂Q are closed curves meeting transversally, I is even. The published state of the art [5] gives $I \leq 40$, and 38 is attained. So the entire question is whether 40 is attainable.

1.2 What is proved here

Theorem 1.2 (Main, conditional)**.** *Let P and Q be simple heptagons in the plane, each simple as a point set, such that no vertex of either lies on the boundary of the other. Then*

$$|\partial P \cap \partial Q| \;\leq\; 38,$$

and 38 *is attained. No condition on collinearity of vertices is assumed.*

The hypothesis "no vertex of either lies on the boundary of the other" is exactly the conclusion of Ackerman–Keszegh–Rote [2, §9]. We do not prove it; §8 states precisely what we import and why, and §8.1 reports an unsuccessful attempt to remove it.

Remark 1.3. Theorem 1.2 is **not** $f(7,7) = 38$. Wherever that equation appears in this paper it is written conditionally, on Proposition 8.3. Theorem 1.2 is the statement one obtains after the general-position reduction has been applied, and that reduction is imported, not proved.

1.3 What is new

The contribution is a reduction, not a technique. Both ingredients we use — encoding realizability of a crossing pattern as an oriented-matroid satisfiability problem, and refuting it mechanically — are standard; see §10. What appears to be new is:

- **Two patterns suffice** (Theorem 4.1). A search that was 13,003,620 cases becomes two non-realizability facts plus a degree argument over 28 cases.
- **The smaller pattern is human-readable** and needs no simplicity hypothesis: it constrains three consecutive sides of each polygon and prescribes eight incidences (§3).
- **The reduction to the plane is unconditional.** The combinatorial argument is about orientation signs; getting from there to genuine point sets in $\mathbb{R}^2$ — including removing the collinearity assumption that [2, §3] assume "for convenience" and say can "be achieved easily", without proof (quoted in full in §8) — is carried out in full (§7).

A bounded prior-art search found no previous statement of the covering result. We stress in §10 that this is weak evidence and not a novelty claim.

2 Preliminaries

2.1 Configurations and the disjointness graph

Let P have vertices $p_0, \ldots, p_6$ and Q have $q_0, \ldots, q_6$, indices mod 7, with sides $P_i = [p_i, p_{i+1}]$ and $Q_j = [q_j, q_{j+1}]$. For points a, b, c write

$$\mathrm{or}(a,b,c) = (b_1 - a_1)(c_2 - a_2) - (b_2 - a_2)(c_1 - a_1),$$

twice the signed area. Sides $[a,b]$ and $[c,d]$ *cross* when

$$\mathrm{or}(a,b,c)\cdot\mathrm{or}(a,b,d)<0 \quad\text{and}\quad \mathrm{or}(c,d,a)\cdot\mathrm{or}(c,d,b)<0.$$

This is a conjunction of *strict* inequalities, a fact used repeatedly: it makes crossing an open condition.

Definition 2.1 (after [2])**.** The *disjointness graph* G^D, whose name and notation we take from [2], is the bipartite graph on $\{P_0,\dots,P_6\}\sqcup\{Q_0,\dots,Q_6\}$ with an edge P_iQ_j when P_i and Q_j do *not* cross.

In general position each side pair contributes at most one intersection point, and exactly one when it crosses, so writing I for the number of intersection points and $|E^D|$ for the number of edges of G^D,

$$I+|E^D|=49. \tag{1}$$

2.2 Two unconditional bounds

Lemma 2.2. *I is even.*

Proof sketch. ∂P and ∂Q are closed curves meeting transversally, so their mod-2 intersection number is 0: each side of P contributes to I once per transversal crossing, and summing the crossing parities around the closed walk ∂P returns to the starting value. Hence $I\equiv 0 \pmod 2$. The formal argument is carried out on the orientation data, so the Jordan curve theorem is not used — neither here nor anywhere else in this paper. □

Lemma 2.3. *$\delta(G^D)\geq 1$, and consequently $I\leq 42$.*

Proof. Fix a side P_i and let ℓ be the line through it. Since ∂Q is a closed curve, ℓ meets it an even number of times, so ℓ cannot cross all seven sides of Q; in particular P_i fails to cross at least one Q_j. The same holds with the roles exchanged, so every one of the fourteen vertices of G^D has degree at least 1 and $|E^D|\geq 7$. With (1) this gives $I\leq 42$. □

Remark 2.4 (Where simplicity is and is not needed)*.* Lemma 2.3 does *not* need simplicity, and it is worth being precise about why, because the neighbouring corollaries do.

The argument above uses only that ∂Q is a *closed* walk, which is true whether or not it crosses itself. So $\delta(G^D)\geq 1$ and $I\leq 42$ hold for any closed polygons in general position.

What does need simplicity is the *exclusion* of 40 and 42, because Theorem 3.6's seven self-non-crossing conditions are supplied by it. Here is a witness, and note carefully what it does and does not show: the 14 points

$$P:(197,86),(182,258),(152,128),(269,227),(90,125),(278,210),(132,227),$$
$$Q:(149,267),(228,177),(105,155),(278,223),(38,63),(253,284),(184,98)$$

have all 364 orientation determinants nonzero and satisfy $I=42$; both heptagons are self-crossing, and G^D is a perfect matching. This does *not* contradict Lemma 2.3 — it *attains* the bound. What it contradicts is Corollary 5.1 read without a simplicity hypothesis, which is why that hypothesis appears there and in Corollary 5.2. Lemma 2.2, like Lemma 2.3, needs no simplicity at all.

Remark 2.5. Černý et al. [5, Lemma 1] obtain $\delta(G^D)\geq 1$ from the fact that a line meets a simple 7-gon in at most 6 sides. We do not import that: the minimum-degree statement is derived directly and is machine-checked unconditionally, so their Lemma 1 is corroboration here, not a dependency.

By Lemmas 2.2 and 2.3, $I\in\{\dots,38,40,42\}$, and since 38 is attained (§6) the theorem reduces to excluding 40 and 42.

2.3 Orientation data

All of §§3–5 concerns only the signs of the $\binom{14}{3} = 364$ determinants $\mathrm{or}(v_a, v_b, v_c)$ on the combined vertex set. A *configuration* is a 14-tuple of points together with the requirement that all 364 signs be nonzero; a *crossing pattern* is the resulting 7×7 Boolean matrix of side-pair crossings. Realizability questions about patterns are therefore questions about which sign vectors arise from actual point sets — that is, about chirotopes.

3 Two forbidden patterns

The plan for the next two sections: we exhibit two local crossing patterns and show in §3.4 that neither is realizable; then §4 shows that every crossing pattern with $I \in \{40, 42\}$ contains one of them. The two patterns were *found* by computer search — minimising unsatisfiable cores — but what they are is small enough to state in a table, and Π_8 is small enough to check by hand.

3.1 The small pattern

Definition 3.1. Pattern Π_8 constrains sides P_0, P_1, P_2 and Q_0, Q_1, Q_2:

	Q_0	Q_1	Q_2
P_0	$\times$	$\times$	$\times$
P_1	$\times$	$\cdot$	$\times$
P_2	$\times$	$\times$	$*$

where $\times$ means *crosses*, $\cdot$ means *does not cross*, and $*$ is unconstrained. That is eight prescribed incidences: seven crossings and one non-crossing.

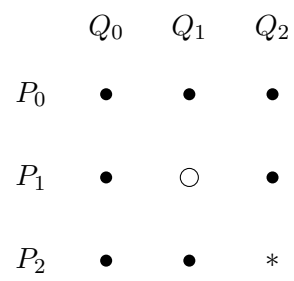


Figure 1: The pattern Π_8. A filled dot means the two sides cross; the open circle at (P_1, Q_1) means they do not; $*$ is unconstrained. Eight prescribed incidences on three consecutive sides of each polygon. No simplicity hypothesis is needed.

Theorem 3.2. *No configuration realizes* Π_8.

Theorem 3.2 assumes nothing beyond the eight incidences and the nondegeneracy of the 364 determinants: in particular it needs *no* hypothesis that P or Q is simple. It is the single most compact *statement* in this work.

We must be equally plain about the other half: **we offer no human-surveyable proof of Theorem 3.2 or of Theorem 3.6.** What is small enough to check by hand is the *pattern*, not its non-realizability. A reader who wants to attack the result should attack Π_8: eight incidences on three consecutive sides of each polygon is a small enough object that a synthetic argument, or a counterexample, may well be findable by a specialist. We could not find either.

What we can offer instead is an independent verification by a completely different route, described next.

3.2 An independent check of Π_8 by order types

Π_8 constrains three consecutive sides of each polygon, so it involves exactly **eight points**, $P_0, \dots, P_3$ and $Q_0, \dots, Q_3$. Whether two segments cross is determined by orientation signs, hence by the *order type* of those eight points — and the order types of eight points in general position are completely enumerated: there are 3315 of them [4, 3]. So Theorem 3.2 can be decided by enumeration, with no solver and no proof assistant.

We did this. Checking every labelling of every one of the 3315 representatives:

no order type of eight points admits Π_8 — 0 of 3315.

Two conventions might have left a gap and neither does. The database identifies mirror images, which is harmless here because reflection negates every orientation while each crossing test is a *product* of two, so the crossing relation is unchanged. And it stores one representative per order type, so the search enumerates labellings itself.

Remark 3.3 (Controls, and a by-product). "We found nothing" is worth nothing without a control that the search can find something. Two were run. The crossing pattern taken from the certificate of §6 *is* located in the database. Sharper: dropping any single one of Π_8's eight conditions yields a pattern that *is* realizable, and all eight relaxations are found. That rules out a vacuously failing search, and as a by-product it confirms Π_8 **is minimal** — a property we had previously established only by randomised minimisation of unsatisfiable cores, i.e. by the very machinery this check is meant to be independent of.

This is corroboration, not a second proof: it is independent of the Lean route in its source of truth (an externally published database), its algorithm (enumeration, not refutation), and its implementation.

Remark 3.4 (The same check cannot be run for Π_{14}). Π_{14} constrains four consecutive sides of P and five of Q, so it involves $5 + 6 = \mathbf{11}$ points, not eight. The enumeration of order types is complete only to $n = 10$ — and already 14,309,547 there — so this route is unavailable for Π_{14} and we do not have an independent exhaustive check of Theorem 3.6. Nor does the minimality of Π_{14} offer a way round: every proper subset of its conditions is realizable, so there is no smaller non-realizable sub-pattern to catch on fewer points.

3.3 The larger pattern

Definition 3.5. Pattern Π_{14} constrains $P_0, \dots, P_3$ and $Q_0, \dots, Q_4$ by fourteen incidences,

	Q_0	Q_1	Q_2	Q_3	Q_4
P_0	$*$	$\times$	$\times$	$*$	$\times$
P_1	$\times$	$\cdot$	$\times$	$\times$	$\times$
P_2	$\times$	$\times$	$*$	$*$	$\times$
P_3	$\times$	$\times$	$\times$	$*$	$*$

together with seven simplicity conditions: $P_0 \not\times P_2$, $P_0 \not\times P_3$, $P_1 \not\times P_3$, $Q_0 \not\times Q_2$, $Q_0 \not\times Q_3$, $Q_1 \not\times Q_3$, $Q_1 \not\times Q_4$.

Theorem 3.6. *No configuration realizes* Π_{14}.

Unlike Π_8, this one genuinely uses that the polygons do not cross themselves; the seven self-non-crossing conditions are part of the hypothesis and are supplied by simplicity.

3.4 How the patterns are proved non-realizable

Both theorems are established by encoding realizability as a Boolean satisfiability problem over the 364 orientation variables — Grassmann–Plücker relations, acyclicity, and the prescribed incidences — and refuting it. The refutation is not trusted:

1. Each clause of the encoding is *derived from the geometry* inside the proof assistant. The Grassmann–Plücker and acyclicity clauses are polynomial identities in the coordinates; the pattern clauses are the definition of crossing unfolded. Nothing about the encoding is assumed.

2. The solver's refutation is checked, then transcribed into an explicit chain of unit-propagation steps. The kernel verifies a chain of implications between propositions; *no search runs inside the proof assistant.*

Remark 3.7 (Controls against a vacuous encoding). The obvious worry is an over-constrained encoding, unsatisfiable for *everything*, which would refute the patterns for the wrong reason. Two things guard against it.

The substantive one: because every clause is *derived* from the geometry inside the proof assistant rather than asserted, a clause that did not in fact follow could not be proved at all. The general-position clauses are theorems about arbitrary configurations and they are satisfiable — the configuration of §6 has all 364 determinants nonzero, checked by exact arithmetic. So unsatisfiability cannot come from the general-position part; it must come from the prescribed incidences.

A cheaper check: the crossing matrix of the 38-crossing configuration of §6 contains *no* relabelled copy of either pattern. Note what has to be checked here — containment in Theorem 4.1 is up to the order-392 group, so observing that the identity labelling fails would establish nothing; all 588 instances must be excluded, and they are. This is a *negative* check that a vacuously unsatisfiable encoding would pass trivially; it is the first argument, not this one, that addresses that failure mode.

Remark 3.8 (Soundness, not completeness, is what is needed). The encoding is a set of *necessary* conditions for realizability. That is the right direction here: we are proving non-realizability, so we need only that every clause is valid. Completeness of the encoding — that every sign vector satisfying the clauses is realizable, which is false in general — is neither claimed nor used.

4 The covering theorem

Theorem 4.1. *Every crossing pattern of two simple heptagons with $I = 40$, and every one with $I = 42$, contains Π_8 or Π_{14} after relabelling.*

Two definitions are needed to read this statement.

A configuration *realizes* a pattern if its crossing relation agrees with every prescribed entry; entries marked $*$ are unconstrained. A crossing pattern *contains* Π if some relabelling carries Π onto a block of cyclically consecutive side indices of each polygon, on which all the prescribed entries hold.

"Relabelling" is the group of order $392 = 14 \cdot 14 \cdot 2$ generated by rotating and reflecting each polygon independently and exchanging P with Q (legitimate only because $m = n$). On *side* indices a vertex rotation by k sends P_i to P_{i+k}, whereas the reflection $p_i \mapsto p_{-i}$ sends P_i to P_{-i-1}; the shift matters when checking a containment by hand.

4.1 The $I = 42$ case

By (1), $I = 42$ forces $|E^D| = 7$, and with $\delta(G^D) \geq 1$ on both sides the graph is a perfect matching. There are $7! = 5040$ of these, falling into 28 orbits under the order-392 group. That every one of them contains Π_8 or Π_{14} is *machine-checked*, by a propagation chain of the same kind as Remark 4.2 describes. It is not a hand argument, and the brevity of this paragraph should not suggest otherwise. (The coincidence that 28 also counts the degree vectors of §4 below is exactly that — the two have nothing to do with each other.)

4.2 The $I = 40$ case, and its 28 degree shapes

Here $|E^D| = 9$, so G^D has nine edges with all fourteen degrees at least 1. Write $r_i = \deg P_i$, so $\sum_i r_i = 9$ with every $r_i \geq 1$. Subtracting one from each, the excess is 2 distributed over seven rows, so either

- one row has degree 3 and the rest degree 1 — 7 placements; or
- two rows have degree 2 and the rest degree 1 — $\binom{7}{2} = 21$ placements, which split by the cyclic distance 1, 2 or 3 between the two rows into $7 + 7 + 7$.

That is $7 + 21 = \mathbf{28}$ degree vectors, and up to rotation of P they are **four** shapes. The same applies to columns, and the column degrees enter through the constraint $\sum_j c_j = 9$ with $c_j \geq 1$.

Remark 4.2 (What is and is not human-sized here). The enumeration above is human-checkable and we have just done it. **The discharge of each of the 28 cases is not.** Each case is settled by a machine propagation chain in the same style as §3.4, drawing on a library of 588 pattern-avoidance lemmas together with the row- and column-degree facts; the four chains total 10,597 propagation steps. We state this explicitly because an earlier draft of this paper described each case as closed by "a short argument", and that was wrong.

Those 588 **lemmas are not additional inputs.** Each is a *relabelled copy* of Π_8 or Π_{14} under the order-392 group, and the count decomposes as $588 = 196 + 392$: the incidence matrix of Π_8 is symmetric under transposition, so the $P \leftrightarrow Q$ swap contributes nothing new for it, while Π_{14} gives the full 392. So "two patterns suffice" survives intact: the library is the two theorems of §3 instantiated, not new mathematics.

So Theorem 4.1 should be read as machine-checked, with the 28-fold case split being the part a reader can survey. What the reduction buys is not that the whole argument became hand-checkable, but that its *inputs* did: two pattern non-realizability facts, one of which is Definition 3.1.

5 Excluding 40 and 42

Corollary 5.1. *No configuration of two simple heptagons has $I = 40$ or $I = 42$.*

Proof. Immediate from Theorems 3.2, 3.6 and 4.1. □

Corollary 5.2. *Every configuration of two simple heptagons has $I \leq 38$.*

Proof. $I \leq 42$ by Lemma 2.3, I is even by Lemma 2.2, and 40 and 42 are excluded by Corollary 5.1. □

6 The lower bound

Proposition 6.1. *The heptagons with vertices*

$$P : (373, 157),\ (132, 5),\ (369, 182),\ (88, 157),\ (327, 282),\ (60, 272),\ (372, 298),$$
$$Q : (344, 24),\ (89, 303),\ (345, 19),\ (373, 214),\ (270, 338),\ (325, 93),\ (192, 284)$$

are simple, are in general position, and their boundaries meet in exactly 38 *points.*

All three assertions are decided by exact integer arithmetic and verified by the kernel. Combined with Corollary 5.2 this gives the equality in Theorem 1.2.

7 From orientation signs to the plane

Sections 3–5 are about sign vectors. Three further steps are needed to reach Theorem 1.2, and all three are carried out unconditionally.

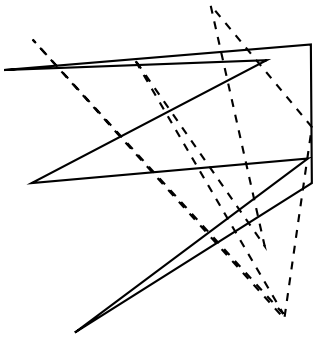


Figure 2: The configuration of Proposition 6.1: P solid, Q dashed. Their boundaries meet in 38 points. Drawn to scale from the integer coordinates.

7.1 Integer coordinates are no restriction

The combinatorial development uses integer coordinates. This costs nothing. Fix a configuration V with all 364 signs nonzero. Each sign condition is a strict inequality in the 28 coordinates, so the set of tuples with the same sign vector is open and contains a product of coordinate intervals around V, say of width $\delta > 0$. Since or is homogeneous of degree 2 that set is also scale-invariant, so tV is good for every $t > 0$ and the corresponding product of intervals around tV has width $t\delta$. Choose $t > 1/\delta$: every coordinate interval then has length exceeding 1 and so contains an integer, and rounding each coordinate independently lands on an integer point with the same 364 signs. No rationals and no clearing of denominators are required.

7.2 Counting points, not side pairs

The combinatorial argument counts crossing *pairs*; the theorem counts *points*. Under Hypothesis 8.1 — *not* under full general position, which Theorem 1.2 does not assume — these agree: a crossing pair meets in exactly one point, and conversely a shared point lying on a pair with a vanishing determinant would put a vertex on the other boundary, which (E9) forbids. The chain is then

$$\#\text{points}(V) \;=\; \#\text{pairs}(V) \;\le\; \#\text{pairs}(V') \;\le\; 38,$$

where V' is a perturbation as in §7.3. Both equalities and both inequalities are proved.

7.3 Removing the collinearity assumption

[2, §3] assumes, "for convenience" and without proof, that no three of the combined vertices are collinear. We do not assume it. The argument is a perturbation, in two halves:

Lemma 7.1. *Crossing is an open condition, so for a small enough perturbation of the fourteen vertices no existing crossing is lost.*

Lemma 7.2. *If two segments are disjoint, they remain disjoint when their four endpoints move by less than some $\varepsilon > 0$.*

Lemma 7.1 handles the crossings; Lemma 7.2 handles simplicity. These need *different* arguments: for a *given* pair of sides, "does not cross" is a closed condition, so it cannot be inherited from the openness of crossing that gives Lemma 7.1. Simplicity is nonetheless stable, because for a polygon simple as a point set the non-adjacent closed sides are disjoint compact sets and hence at positive distance — which is what Lemma 7.2 exploits. The configurations in general position are dense: the complement is the zero set of finitely many nonzero polynomials in the 28 coordinates, hence a finite union of hypersurfaces. Combining this with the two lemmas approximates an arbitrary pair of point-set simple heptagons by one in general position with

at least as many crossings, giving the bound for the original. A single ε must serve every crossing, every disjoint side pair and all 364 determinants at once; since there are finitely many conditions, one takes the minimum.

One case deserves separate mention, since Lemma 7.2 does not cover it: *adjacent* sides share an endpoint and so are never disjoint. Simplicity constrains them too — they must meet only at that shared vertex — and this is exactly the delicate case, because three consecutive collinear vertices are what §7.3 exists to permit. In the formal development the adjacent case is handled separately, by the observation that once the perturbed configuration is in general position its adjacent sides cannot overlap. Lemma 7.2 itself is proved for an arbitrary real normed space and is independent of everything else here.

Remark 7.3. Note the direction: we need the perturbed configuration to have *at least* as many crossings, not the same number, and openness gives exactly that.

8 What is imported

Hypothesis 8.1 ((E9))**.** No vertex of either polygon lies on the boundary of the other.

This is the conclusion of [2, §9], where §9.1 removes vertex–vertex coincidences and §9.2 vertex–side ones.

Remark 8.2 (What is conditional, exactly). Theorem 1.2 is itself *unconditional*: like any theorem it has hypotheses, and Hypothesis 8.1 is one of them. What is conditional is the *deduction* of $f(7,7) = 38$, and that needs a *reduction*, not a property. We state it with explicit quantifiers, since this is the precise object we import.

Proposition 8.3 (Imported; [2, §9])**.** *For every pair (P, Q) of simple heptagons with finitely many boundary intersections there is a pair (P', Q') of simple heptagons satisfying Hypothesis 8.1 with $|\partial P' \cap \partial Q'| \geq |\partial P \cap \partial Q|$.*

Corollary 8.4. *Proposition 8.3 together with Theorem 1.2 gives $f(7,7) = 38$.*

Proof. Given any pair, Proposition 8.3 produces one satisfying Hypothesis 8.1 with at least as many intersection points, and Theorem 1.2 bounds that by 38; the bound is attained by Proposition 6.1. □

Proposition 8.3 is the *only* thing we take on trust, and we do not prove it. Three things should be said about it plainly.

Remark 8.5 (Provenance). We read [2] in its journal version, which is the only version containing §9: the SoCG/arXiv version [1] ends at §8 and contains no treatment of degeneracies at all. A referee checking our citation against arXiv will not find the section.

Remark 8.6 (What §9 says, in its own words). Since we ask the reader to accept that Hypothesis 8.1 is what §9 delivers, we quote it. [2, §9] opens:

> "We will now justify our general-position assumption that we introduced in Sect. 3 stating that *no vertex of one polygon lies on the boundary of the other polygon.* That is, we will show that one can get rid of such degenerate cases *without decreasing the number of intersection points*."

The second clause is exactly the inequality of Proposition 8.3: the Proposition is their sentence, not our paraphrase of it.

Remark 8.7 (Their standing assumption is broader, and why that is harmless). [2] assume throughout that no three of the combined vertices are collinear *and* that no two of the combined sides are parallel. We use neither. One might worry that §9's own argument runs under that full assumption, in which case Proposition 8.3 would be stronger than what §9 delivers, and §7.3 would buy nothing at the level of the final claim.

It does not, and their §3 is explicit about the order. It first states the vertex condition, "justified in Sect. 9 by a non-trivial perturbation"; and only then, *given* that condition,

> "we assume *for convenience* that no three vertices of P and Q combined lie on a line, and no two sides of P and Q combined are parallel. Indeed, given that no vertex of one polygon lies on the other polygon, this can be achieved easily by another small perturbation of the sides without changing the intersection pattern."

So §9 is *prior* to the collinearity and non-parallelism clauses rather than dependent on them, and §7.3 does remove the former from the final claim. The latter we never use.

The quotation also shows that this second perturbation is **asserted and not proved**: "for convenience", "can be achieved easily", no lemma number and no argument. That is the clause §7.3 discharges, and it is why we treat it separately from §9 rather than lumping both under "general position".

Remark 8.8 (The import is slightly more than §9 states). Theorem 1.2 assumes *both* that the polygons are simple as point sets *and* Hypothesis 8.1, so the reduction must deliver both. That the perturbed polygons remain *simple* is, as far as we can tell, not stated in [2]. What we import is therefore §9's conclusion *together with* simplicity-preservation, and the latter has no published warrant we can point to. We flag it rather than absorb it silently: it is the joint at which a previous attempt on this problem failed.

Note that §9 and §10 appear only in the journal version; the arXiv posting is the SOCG version [1] and does not contain them. It is the *only* external mathematical input to Theorem 1.2.

Why it is needed, precisely: without it, a point of $\partial P \cap \partial Q$ can lie on *no* crossing pair. Two sides that merely touch share a point but have vanishing determinants, so the pair is not counted, and the bound on pairs does not bound points. The failure is in the unfavourable direction, which is why §7.3 does not also dispose of it.

Remark 8.9 (Citation). It would be an under-citation to attribute all of the usual general-position package to §9. §9 gives Hypothesis 8.1 and nothing more; the collinearity clause has its only warrant in one unproved sentence of §3, and is discharged here (§7.3). Additionally, that the perturbed polygons remain simple is stated nowhere in that paper. We fold simplicity into Hypothesis 8.1 deliberately, so that the gap is visible.

8.1 An attempt to remove it

We attempted to prove Hypothesis 8.1 and did not succeed. The decomposition into its vertex–vertex and vertex–side halves is formalized, as are two local lemmas, but the global reduction is not. Two negative observations came out of the attempt and are worth recording, because they close off the obvious routes:

- Generic perturbation preserving proper crossings does *not* imply that the point count is monotone.
- Naive displacement into the interior of a cone does not either.

Explicit coordinates exhibiting both failures are recorded in the project repository; they are analytic and *not* machine-checked. This is consistent with the case on which an earlier attempt of ours broke: writing $\mathrm{Cone}_P(v)$ for the cone spanned at a vertex v by its two incident sides,

the difficulty arises when $\mathrm{Cone}_P(v) \subseteq \mathrm{Cone}_Q(w)$, since then displacing v into the cone at w can destroy the intersection point rather than split it. None of this is evidence that AKR's theorem is false, and we allege no error in it; what remains is substantial unfinished work.

9 Machine verification, and its limits

Every mathematical assertion in §§2–7 is verified by the Lean 4 kernel. There is no `sorry`, no `native_decide`, and no project-specific axiom; every theorem depends on at most the three standard axioms `propext`, `Classical.choice`, `Quot.sound`. Hypothesis 8.1 is a hypothesis of the statement, never an axiom.

Remark 9.1 (What the kernel does not check). The kernel proves theorems about definitions; it does not certify that the definitions mean what we say. That $\partial P \cap \partial Q$ is modelled by the right set, and that "simple heptagon" is modelled by the right predicate, are judgements a human reader must make. This is, in our view, the principal remaining risk, and it is not reducible by more computation.

One partial check is available and we have carried it out: the definitions, applied to the explicit 38-crossing configuration of Proposition 6.1, yield exactly 38. A wrong definition would have to be wrong in a way that still lands on that value.

9.1 Which results a human can check, and which are machine-only

We think a referee is owed this table explicitly rather than having to infer it.

Result	**Status**	**Checkable by hand?**
$I + \lvert E^D\rvert = 49$, Lemma 2.2, Lemma 2.3	machine-checked	**yes**, and short
The 28 degree shapes (§4)	machine-checked	**yes**; derived above
Theorem 3.2 (Π_8 non-realizable)	machine-checked	**no** — but see §3.2
Theorem 3.6 (Π_{14})	machine-checked	**no** — solver refutation
Theorem 4.1 (covering)	machine-checked	**no** — see Rem. 4.2
Proposition 6.1 (the 38 example)	machine-checked	**yes** in principle; 364 determinants
§7 (reduction to the plane)	machine-checked	**yes**; ordinary analysis
Hypothesis 8.1	**imported, not proved**	it is [2, §9]

The three "no" rows are the substance of the exclusion, and we do not want to soften that. What the reduction achieves is that the machine-only part now rests on *two* non-realizability facts about small explicit patterns instead of on an enumeration of 13,003,620 objects — and one of those two, Π_8, is small enough to state in a 3×3 table (Figure 1).

9.2 How to reproduce

The development is at `https://github.com/NewYorkImperialist/heptagon-intersections`, at commit `8e6358a`. With Lean 4.33.1 and its Mathlib pinned by the repository:

`cd lean && lake build`	builds everything; no `sorry`
`bash scripts/verify-all.sh`	full check, exits 0
`#print axioms` lines	in every module; expect the three standard

The results of this paper correspond to the following declarations.

In this paper	In the development
Theorem 3.2	`MinPatternClauses000.minpattern000_impossible`
Theorem 3.6	`MinPatternClauses001.minpattern001_impossible`
Theorem 1.2	`AKR9.points_le_38_of_no_vertex_on_boundary`
the equality	`AKR9.isGreatest_38_of_E9`
Lemma 7.2	`SegmentStable.exists_eps_disjoint_stable`
Hypothesis 8.1	`AKR9.NoVertexOnOther`

`SegmentStable` imports only Mathlib, so Lemma 7.2 can be read on its own.

The solver refutations are shipped as certificates and re-checked by a standalone proof checker independent of the proof assistant; that checker is itself first shown to reject non-refutations.

9.3 Independent corroboration

Beyond the kernel: the terminal case analysis of an earlier version of this argument was reproduced by a second, independently written implementation in a different language; all solver refutations are re-checked by a standalone proof checker, which is itself first shown to reject non-refutations; and all generated artifacts are regenerated and compared byte-for-byte. These guard against implementation error. *None of them is a check on whether the argument is the right argument.*

10 Related work

Dillencourt, Mount and Saalfeld [6] posed the problem and settled the even cases. Černý, Kára, Král', Podbrdský, Sotaková and Šámal [5] gave bounds for odd cases and solved $k = 5$, explicitly leaving $k = \ell = 7$ open. Ackerman, Keszegh and Rote [2] obtained asymptotically optimal bounds; their §9 is what we import. Günther's preprint [7] on odd n was withdrawn by its author.

Both of our tools are standard. Encoding order-type or chirotope realizability as a satisfiability problem and refuting it mechanically is well established, including in combination with proof assistants. Framing extremal results in terms of forbidden configurations is likewise standard. What we have not found in the literature is either applied to crossing patterns of two simple polygons, or to $f(m, n)$.

Remark 10.1 (On novelty). We searched for prior statements of the covering result and found none. We record plainly that this is **weak evidence and not a novelty claim**: the search was bounded, the forward citations of [5] were not examined, and the relevant literature is small enough that a result could sit in it under different vocabulary. A domain expert's judgement is not replaced by such a search.

11 Further work: the family $f(7, n)$

We record an observation which suggests that the natural sequel to this work is not $f(9, 9)$ but the whole family $f(7, n)$, and which costs two lines.

G^D has $m + n$ vertices and, by Lemma 2.3, no isolated ones. A graph on N vertices with c connected components has at least $N - c$ edges, so $|E^D| \geq (m + n) - c$ and, with (1),

$$I \;\leq\; mn - (m + n) + c.$$

For m, n both odd, $mn - (m + n)$ is odd while I is even. Hence $c \leq 4$ already forces $I \leq mn - (m + n) + 3$, the conjectured value: **to prove the conjecture for a given (m, n) it suffices to bound the number of components of G^D by 4.**

Now fix $m = 7$. Every component of G^D contains an edge and therefore a side of P, and there are only seven of those, so $c \leq 7$ *automatically, for every* n. Consequently

$$f(7,n) = 6n - 4 \text{ for all odd } n \quad \text{would follow from ruling out } c \in \{5,6,7\},$$

three cases whose number does not grow with n. Theorems 3.2, 3.6 and 4.1 of this paper do exactly that for $n = 7$: $I = 42$ forces $c \geq 7$ and $I = 40$ forces $c \geq 5$.

This places the problem on a visible ladder. At $m = 5$ only $c = 5$ has to be excluded — a single case — and that is precisely the range [5] settle, obtaining $f(5,\ell) = 4\ell - 2$ for every ℓ. So $m = 7$ is the next rung above a solved problem, at three cases rather than one, and this paper supplies its base case.

Two further remarks, one encouraging and one not.

Remark 11.1 (The patterns are not about heptagons). Neither Definition 3.1 nor Definition 3.5 mentions the number of sides: Π_8 is a statement about eight points and three consecutive sides of each polygon, and Π_{14} about eleven. Theorems 3.2 and 3.6 are therefore already theorems about n-gons for every n, and would be reused unchanged. What does not transfer is Theorem 4.1, the covering claim, which is verified by enumeration over a domain that depends on n.

Remark 11.2 (Why a single pattern cannot suffice, for any fixed n-free argument). The case $c = 7$ is rigid: each component holds exactly one side of P, so every side of Q fails to cross *exactly one* side of P, giving a surjection σ from the sides of Q onto those of P. Suppose $\sigma(j) \equiv j \pmod 7$. Then in any window of three consecutive sides of each polygon, the neighbouring non-crossing already falls inside the window, and Π_8 never occurs. So no argument for all n can rest on Π_8 alone; it would need a family of patterns closed under this kind of structure, or a different idea altogether. We do not know whether a *finite* family suffices for all n, and if the number required grows with n then this route fails.

We make no claim that any of this is easy. The point is only that the obstruction is identifiable and the target is a single, finitely stated combinatorial question, and that a general argument — not an enumeration — would be needed, since "all n" is not something a computation reaches.

12 Status, and what we ask of a reader

To state the position exactly:

- $f(7,7) = 38$ is **not** proved. Theorem 1.2 carries Hypothesis 8.1, and the deduction needs Proposition 8.3, which we import.
- Every mathematical assertion in §§2–7 is machine-checked from primitive definitions, with no published result about $f(m,n)$ appealed to.
- **No mathematician outside this process has reviewed any of it.**

12.1 What can be judged without a proof assistant

The third point is the binding constraint and is why this paper exists. We stress that **no familiarity with Lean, or with formal verification at all, is needed to assess the parts where an expert's judgement is irreplaceable.** It is worth being precise about why.

The machine-checked proofs in this work are *not* human-readable and are not meant to be: they are propagation chains of thousands of steps, generated rather than written. Nobody reads them — a formalization specialist would not either. What any reader of a formal development actually inspects is the *statements* and the *definitions*, and we have put both into ordinary mathematical notation, the definitions in Appendix A. So the practical difference between a Lean-literate reader and any other is close to nil.

Three questions are the ones we would most value, and none of them involves a machine:

1. **Is Theorem 3.2 true?** Eight incidences on three consecutive sides of each polygon, no simplicity hypothesis (Definition 3.1). This is a question of plane geometry. We offer no human proof of it, so a synthetic argument — or a counterexample — would be decisive either way. §3.2 gives an independent verification by order-type enumeration that a reader can re-run in seconds without Lean, which is the nearest thing to a hand check we can supply. If it is false, the exclusion of 40 and 42 collapses, though §2, §6 and §7 would survive.

2. **Does [2, §9] actually deliver Proposition 8.3?** This is the one thing we import, and settling it is a matter of reading that paper, not this one. The specific worry we cannot resolve ourselves is recorded in the remark following Proposition 8.3: whether §9's argument depends on AKR's standing non-parallelism assumption.

3. **Is the covering result (Theorem 4.1) already known?** Our search found nothing, and we make no novelty claim (Remark 10.1).

We would also welcome an opinion on Appendix A — whether those definitions are the intended objects — which is the risk Remark 9.1 identifies and the one thing no amount of computation can settle.

Disclosure of generative-AI use

Generative-AI systems were used substantially throughout: exploring the reduction, generating essentially all software including the formalization, auditing the cited sources, and drafting this paper. The named author is responsible for the claims. AI output was not treated as trustworthy: each major claim is discharged by a separately recorded mechanism — source audit, independent reimplementation, a machine-checkable certificate with a standalone checker, or kernel verification — and errors introduced during development are catalogued in the project repository rather than quietly repaired. The patterns of §3 were *found* by computer search; what they are is small enough to check by hand.

A The formal definitions, in ordinary notation

Remark 9.1 identifies definitional faithfulness as the principal residual risk: the kernel proves theorems *about* these definitions and cannot certify that they are the intended objects. That judgement is a reader's, so we set them out here. These are transcriptions of the formal statements, not paraphrases; indices are mod 7 and $[x, y]$ denotes the closed segment.

Let $P, Q : \mathbb{Z}/7 \to \mathbb{R}^2$.

Simple as a point set. P is *simple* if
(S1) $P_i \neq P_{i+1}$ for every i (no degenerate side);
(S2) $[P_i, P_{i+1}] \cap [P_j, P_{j+1}] = \emptyset$ whenever P_i and P_j are non-adjacent sides;
(S3) $[P_i, P_{i+1}] \cap [P_{i+1}, P_{i+2}] = \{P_{i+1}\}$ for every i.
Nothing here constrains collinearity: three consecutive vertices of P may be collinear, and so may any three of the fourteen.

The hypothesis (E9). Hypothesis 8.1 is

$$P_i \notin [Q_j, Q_{j+1}] \quad \text{and} \quad Q_j \notin [P_i, P_{i+1}] \qquad \text{for all } i, j.$$

What is counted. The boundaries and their intersection are

$$\partial P = \bigcup_i [P_i, P_{i+1}], \qquad \partial Q = \bigcup_j [Q_j, Q_{j+1}], \qquad X = \partial P \cap \partial Q,$$

genuine subsets of $\mathbb{R}^2$, and the quantity bounded is the cardinality of X. Finiteness of X is *proved*, not assumed — it follows from the injection of X into the set of crossing side pairs — which matters, because the formal cardinality function returns 0 on infinite sets and a bound stated without finiteness would be vacuous for overlapping boundaries rather than informative.

The theorem. With P, Q simple in the above sense and (E9) holding, $|X| \leq 38$; and 38 is attained, by Proposition 6.1.

The question we are asking is simply whether (S1)–(S3), (E9) and X above are the objects a geometer means by "two simple heptagons whose boundaries meet in k points". If they are not, no amount of machine checking helps.

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